

PORTFOLIO DECISION SYSTEM (PDS)

Architecture, Governance, and Operating Framework

A framework for combining independent strategy targets into a single monitored portfolio

SYSTEM DOCUMENTATION v1.1

This document explains how the Portfolio Decision System is structured, how portfolio decisions are formed, and how the system is operated. It is not an investment recommendation, a research paper on the underlying models, or a record of broker executions.

SlackQuant Research

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research.slackquant.com/systems/pds/

Document Summary

Field	Details
Document	Portfolio Decision System (PDS): Architecture, Governance, and Operating Framework
Version	1.1
Release date	2026-09-11
System type	Portfolio Decision & Operating System
Public scope	System architecture and delayed historical results are public; current portfolio targets and monitoring outputs remain private.
Current Core	F2R + ADAA under a governed strategic allocation, rebalanced monthly. Exact provider composition weights are private.
Decision and rebalance cycle	Monthly signals; rebalance at the first U.S. trading-day close available to both strategies after the signal month ends.
Performance measurement	Performance is calculated from a continuous daily return series; monthly and annual figures are aggregated from it.
Disclosure policy	Published materials use delayed historical data. Current targets, Forward Preview, Forward Shadow, Current Mark, and current FX positioning are not disclosed.

How this document is organized

Sections 1-4 explain what PDS does and how it is structured. Sections 5-9 cover portfolio construction, timing, performance, portfolio status, and FX. Sections 10-12 describe monitoring, operations, and public disclosure. Sections 13-15 cover release validation, related methods, and limitations. The appendices summarize decision status, public resources, and the v1.1 specification.

Terminology

In this document, F2R means Forecast-to-Rank Allocation.

OVERVIEW

1. Executive Overview

Portfolio Decision System (PDS) combines target portfolios from independently developed investment strategies into one portfolio. It does not alter the underlying strategy logic. PDS supplies the common rules needed to combine targets, rebalance the portfolio, measure performance, monitor the current state, and publish delayed historical information.

The current Core integrates Forecast-to-Rank Allocation (F2R) and ADAA under a governed strategic allocation. Exact provider composition weights are not part of the public disclosure layer, while the same current-system policy is applied consistently to the historical PDS Core shown in public materials. The portfolio is rebalanced monthly; between rebalances, provider and asset weights move with market returns rather than being continuously reset to strategic targets.



A useful distinction

F2R and ADAA strategy targets, the PDS Official Decision, Forward Preview, Forward Shadow, and published history serve different purposes. Only the Official Decision defines the current PDS portfolio target.

What PDS does

- Combines strategy targets into one portfolio using a consistent set of portfolio rules.
- Applies the same rebalance, turnover, cost, and performance rules when the strategies are evaluated together.
- Keeps the current Official Decision separate from Forward Preview and Forward Shadow outputs.
- Retains the last valid Official Decision when required data or timing checks cannot be reconciled.
- Publishes only delayed historical information that has passed the release checks.

What PDS does not do

- It does not redesign the investment logic of F2R or ADAA.
- It does not treat Forward Preview, Forward Shadow, or FX research as an approved portfolio decision.
- An Official Decision records the PDS target; it is not a broker fill or an account-level trade confirmation.
- Public PDS materials do not disclose or replace the current private portfolio state.

ARCHITECTURE

2. System Architecture

The seven steps below show how a strategy target moves from receipt to the Official portfolio and then into monitoring. PDS standardizes target format, input checks, portfolio combination, rebalance timing, performance measurement, and monitoring.

Stage	Step	What happens
01	Strategy targets	Each strategy produces a target portfolio according to its own rules.
02	Common format	Targets are put into a consistent asset and weight format so they can be combined without changing the underlying strategy rules.
03	Input checks	Strategy identity, date, timing, completeness, and weight consistency are checked before use.
04	Strategy review	Research evidence and practical portfolio value are reviewed before a strategy is approved for a specific role.
05	Portfolio combination	Approved targets are combined using the current PDS allocation policy.
06	Portfolio decision	The monthly PDS target is set and rebalanced under the common decision and execution schedule.
07	Monitoring and review	Realized performance, strategy availability, Forward Preview and Forward Shadow outputs, and any need for further review are monitored.

What PDS standardizes

PDS does not recreate a strategy output when an input is missing. It uses only targets that can be traced to the underlying strategy and that pass the required date, timing, and completeness checks. If a required input cannot be reconciled, that update does not replace the current Official Decision; the last valid official state remains in place.

Independent strategies, common portfolio rules

F2R and ADAA retain their own investment logic. PDS standardizes only the elements that must be common at portfolio level: target format, timing, portfolio combination, performance measurement, and monitoring.

How the components fit together

System	Role in PDS	What remains separate
F2R	Produces a machine-learning cross-asset target portfolio used by PDS.	F2R retains its forecasting, ranking, and target-construction rules.
ADAA	Produces a multi-asset target portfolio used by PDS.	ADAA retains its allocation rules and target construction.
PDS	Combines the strategy targets and applies the common portfolio rules.	PDS does not recreate or override the underlying strategy models.

STRATEGY INPUTS

3. Strategy Inputs and Responsibilities

F2R and ADAA remain responsible for their own calculations. PDS uses their dated target portfolios as inputs, checks that the strategy identity, date, timing, and weights are consistent, and then combines them under PDS rules. It does not fill in a missing strategy output by recreating the strategy inside PDS.

Checks before a target is used

- The correct strategy, target date, and relevant version are identified.
- Signal month, holding month, market-data cutoff, and rebalance timing line up.
- Weights are complete, properly scaled, and traceable to the strategy output.
- Strategy-level costs and PDS rebalancing costs are kept separate so the same cost is not counted twice.
- Missing, stale, or conflicting outputs are not replaced by estimates; the last valid Official Decision remains in place.

Research history and PDS operating history

A strategy's original research record may use a different execution or cost convention from the PDS operating history. PDS keeps that record intact. When strategies are compared inside PDS, the common PDS timing and accounting rules are applied to a separate operating history.

Record	Purpose	How it is used
Original strategy research	Preserves the strategy's original backtest, timing, and cost assumptions.	Used to interpret claims about the strategy itself.
PDS operating history	Measures the approved strategies under the same PDS timing and accounting rules.	Used to evaluate the combined portfolio under the PDS timing and accounting rules.
Public historical record	Provides delayed information for completed periods.	Used to communicate completed PDS history to readers.

Historical records are kept separate

Later changes to PDS timing or accounting do not rewrite the original strategy research record. The strategy research, PDS operating history, and public historical record remain separately identifiable.

CORE PORTFOLIO

4. Current Core Construction

The current PDS Core integrates the two strategy target portfolios under a governed strategic allocation. Each strategy first determines its own asset weights. PDS then combines those target portfolios under its governed integration policy after the required input and timing checks are complete. Exact provider composition weights and the integration formula are not part of the public disclosure layer.

PDS target = governed integration of the F2R target and ADAA target

Total target weight = 100%

Exact provider composition weights and the integration formula are private.

The governed provider allocation is set at the monthly rebalance. It is not maintained as a daily constant-weight allocation. Once the rebalance is completed, both strategy weights and underlying asset weights move with market returns until the next rebalance.

Target and portfolio weights

Measure	Meaning	Use
Official target	Portfolio weights set for the next holding period at the monthly rebalance.	Reference for the scheduled rebalance.
Post-rebalance weights	Portfolio weights immediately after the scheduled rebalance.	Starting weights for the next holding period.
Pre-rebalance weights	Portfolio weights immediately before the next rebalance after market movements.	Used to calculate turnover and rebalancing costs.
Current Mark	Provisional valuation of the incomplete holding period.	Monitoring only; does not change completed history or the Official Decision.

Why weight drift matters

If provider weights were reset to their strategic targets every day, it would be a different strategy: exposures, turnover, and returns would all change. PDS therefore allows weights to drift between monthly rebalances and restores the governed strategic allocation only at the next scheduled rebalance.

Key point

The governed provider allocation is set at the monthly rebalance and then allowed to drift; it is not a daily constant-weight allocation.

TIMING

5. Decision and Rebalance Timing

PDS follows a monthly decision cycle. Once the signal month ends, a new target is set using the information available at that point. The portfolio is rebalanced at the first U.S. trading-day close available to both F2R and ADAA. Returns through that close belong to the previously held portfolio; the new weights begin earning from the next close-to-close interval.



How the monthly cycle works

- The completed signal month determines the target for the next holding period.
- The portfolio is not rebalanced at the same month-end close that produced the signal.
- The existing portfolio remains in place through the rebalance close.
- The new weights apply from the next close-to-close return interval.
- Monthly performance is labeled by the holding month in which the return is realized.
- Calendar-year performance is assigned to the year in which the holding-period returns occur.

When dates do not line up

Strategy targets are combined only when their signal dates and relevant market-data dates line up. If one strategy is missing or not yet available for the required date, PDS does not estimate the missing target. The current Official Decision remains in place until the inputs can be reconciled.

Why timing matters

Timing determines which portfolio earns each return. Even a one-day shift can change measured performance and turnover, so the timing rules are part of the portfolio definition rather than a reporting choice.

PERFORMANCE MEASUREMENT

6. Performance Measurement and Transaction Costs

PDS performance is calculated from one continuous daily return series. Monthly, year-to-date, annual, cumulative, drawdown, volatility, Sharpe, and Calmar statistics are all derived from the same series. This avoids inconsistencies that can arise when daily and monthly figures are built separately.

$$\begin{aligned}\text{Daily wealth: } W_d &= W_{(d-1)} \times (1 + r_{d,\text{net}}) \\ \text{Monthly return: } R_{\text{month}} &= W_{\text{month-end}} / W_{\text{prior-month-end}} - 1\end{aligned}$$

Return and risk calculations

Daily returns are the base frequency. Annualized volatility and Sharpe use a 252-trading-day convention. Drawdown is measured from the running peak of the daily wealth series.

Turnover and rebalancing cost

Turnover is calculated from the portfolio weights immediately before the rebalance and the new target weights. PDS defines one-way turnover as one half of the sum of the absolute weight changes.

$$\begin{aligned}\text{One-way turnover} &= 0.5 \times \sum_i | \text{target weight}_i - \text{pre-rebalance weight}_i | \\ \text{PDS rebalancing cost} &= 2 \times \text{one-way turnover} \times 0.001\end{aligned}$$

The current cost assumption is 10 basis points per side. A complete replacement of the portfolio (100% one-way turnover) therefore implies a 20 basis point round-trip cost.

Avoiding double counting

Transaction costs already reflected in the F2R or ADAA return series are not charged again at PDS level. PDS adds only the incremental cost of rebalancing the F2R/ADAA sleeve weights back to the PDS target.

Consistency check

All published PDS return figures must follow the same timing, turnover, cost, and daily performance rules. If two views do not reconcile, the difference is investigated before release.

PORTFOLIO STATUS

7. Official Decision and Monitoring Views

The PDS dashboard shows several views of the portfolio, but they are not interchangeable. The Official Decision is the portfolio target used for the current holding period. Current Mark, Forward Preview, and Forward Shadow are monitoring views, while public releases are delayed records of completed periods.

View	What it shows	How it is used
Official Decision	The target portfolio in effect for the current holding period.	Used as the PDS portfolio target. It is not a broker fill or trade confirmation.
Current Mark	A provisional view of performance for the current, incomplete holding period.	Monitoring only; it does not change the Official Decision or completed performance history.
Forward Preview	What the next target would look like if the current month ended using the latest available data.	Monitoring only; it is not executed and no Forward Preview performance is reported.
Forward Shadow	A separately tracked alternative to the Active Core used for comparison.	Not traded as part of the Active Core; any performance is kept separate.
Public Historical Release	A delayed record of completed periods released for readers.	Historical information for completed periods.

More recent does not mean official

A newer Forward Preview or Forward Shadow result does not replace the current Official Decision. Public materials can also be recent enough to inform readers while remaining intentionally delayed relative to the private portfolio state.

Changing the Official Decision

A monitoring view becomes part of the Official portfolio only through the regular monthly decision process. Recency alone does not change its status.

KRW INVESTOR CONTEXT

8. Currency Exposure and the FX Overlay

For a KRW-based investor holding U.S.-listed multi-asset ETFs, exchange-rate movements can materially affect portfolio returns and risk. PDS therefore considers currency exposure at the overall portfolio level rather than embedding a separate currency decision inside one underlying strategy.

Two FX views

FX view	Status	How to interpret it
Historical Dynamic FX comparison	Historical comparison	Shows the delayed historical Dynamic FX comparison with the 5 bp cost assumption used in the public series. It does not show the current FX position.
Implementable FX overlay	Not yet treated as implementable	Would require tradable forward/NDF data, actual availability dates, contract-roll rules, hedge turnover, and transaction costs before its performance could be treated as implementable.

The public PDS materials therefore treat Dynamic FX as a historical comparison, not as evidence of a currently executable hedging program. The current FX position and current overlay decision are not disclosed.

Why the distinction matters

- Spot-FX sensitivity is not the same as realized forward/NDF hedging performance.
- Forward carry, basis, contract rolls, and changes in hedge ratios can affect actual hedging results.
- An FX series developed in a strategy research project is not automatically adopted by PDS.
- Historical FX analysis should not be read as the current hedge position.

GOVERNANCE

9. Research Evidence, Portfolio Usefulness, and Adoption

PDS treats three questions separately: whether the research evidence supports a claim, whether a strategy is useful for a particular portfolio role, and whether it has actually been adopted into the Official portfolio. A favorable answer to one question does not automatically answer the others.

Review	Question	What it determines
Research review (Gate R)	Does the evidence support the research claim being made?	Whether the research claim passes, fails, or remains inconclusive.
Portfolio-role review (Gate P)	Does the strategy add enough practical value for the portfolio role being tested?	Whether the strategy is suitable for that role, unsuitable, or still inconclusive.
Official adoption	Has the strategy or allocation actually been approved for the PDS portfolio?	Only an explicit adoption decision changes the Official portfolio.

These are separate decisions

A strategy can have sound research evidence without being useful for the role being considered. It can also look useful in a portfolio test without supporting a new research claim. Neither result changes the Official Core unless the strategy is explicitly adopted.

When a review does not pass

If the required data, timing, or evidence cannot be verified, the proposed update is not used. A problem with Forward Preview does not change the last valid Official Decision. A strategy that does not pass the review for a particular portfolio role remains outside the Official Core unless a later review supports adoption.

MONITORING

10. Monitoring Views

PDS monitoring covers the current Official portfolio as well as several supporting views. The dashboard shows realized Core performance, target weights, Forward Preview, Forward Shadow, and FX information. Each view has a defined purpose so that monitoring information is not mistaken for a portfolio decision.

Dashboard views

Dashboard view	What it shows	How to interpret it
PM Cockpit	The current Official portfolio decision and the main changes from the previous decision.	Official portfolio view.
Core Performance	Realized PDS Core performance from the continuous daily return series.	Completed history is kept separate from the provisional Current Mark.
Portfolio Weights	Current strategy targets and the resulting asset-level PDS target.	Target weights are distinguished from weights that have drifted since the last rebalance.
Forward Shadow	Performance and allocations for a separately tracked alternative to the Active Core.	Comparison only; it does not become the Active Core because it is displayed or performs well.
Forward Preview	The target that would be indicated if the current month ended using the latest available data.	Monitoring only; it is not executed and no Forward Preview performance is reported.
FX Overlay / FX Performance	Current FX monitoring and delayed historical analysis for the KRW-based investor view.	Historical FX analysis is kept separate from the current FX position.

When data are unavailable

If a strategy update, Forward Preview data, timing reference, or Current Mark is unavailable, PDS keeps the last valid Official information and shows that the newer item is unavailable. Missing data are not estimated simply to keep the dashboard complete.

OPERATIONS AND PUBLICATION

11. Operating and Publication Process

PDS decisions are produced and reviewed in the private operating environment. The public SlackQuant site receives only delayed information that has passed the required checks. The website is a publication channel; it is not the source of portfolio decisions.



Main operating steps

Step	Purpose
Refresh and rebuild	Update strategy inputs and rebuild the PDS dashboard. If a monitoring input is unavailable, the last valid Official Decision remains unchanged.
Review the current state	Check the Official Decision, Current Mark, Forward Preview, Forward Shadow, and related changes before any release is prepared.
Prepare the public update	Create the delayed historical release and verify that current portfolio information is excluded.
Approve and publish	Review the changes, confirm the published figures, and release the update to the SlackQuant site.

Publication checks

- 1. Check the PDS public figures, labels, links, and disclosure boundaries.
- 2. Build the public pages and inspect the rendered result.
- 3. Confirm that the downloadable and linked resources match the approved release.
- 4. Publish only after the checks above pass.

Final review remains explicit

Automation can update data, run checks, and prepare a release, but a substantive public update is reviewed before it is published.

PUBLIC DISCLOSURE

12. Public Disclosure

Public PDS materials are intentionally delayed relative to the private operating system. They are designed to explain the system and show completed historical information without revealing the current portfolio state.

Published	Not published
System architecture and the relationship between PDS, F2R, and ADAA	Current asset-level portfolio targets
Current Core policy: F2R + ADAA under a governed strategic allocation; exact provider composition weights are private.	Forward Preview and current Forward Shadow outputs
Delayed historical targets and performance	Current Mark and current FX position
Delayed historical Dynamic FX comparison	Account-level holdings or trade information
Research, Quantitative Methods, and versioned system documentation	Private operating diagnostics that are not needed to interpret the public system

How public performance is dated

Monthly performance is labeled by the month in which the portfolio return is realized, not by the month in which the signal was formed. Annual figures follow the same holding-period convention.

Status of public materials

Public pages and downloadable files are historical and explanatory materials. They are not trading instructions and do not represent the current private portfolio.

Publication rule

A release is not published if it would reveal current portfolio information or if its figures do not reconcile with the current Core definition, timing rules, and performance methodology.

RELEASE CHECKS

13. Release Validation

Before v1.1 was released, the current Core definition, timing, transaction-cost treatment, performance history, disclosure boundary, and dashboard labels were checked against the operating records. The table summarizes the main checks. They document the state of the system at release; they are not a forecast of future performance.

Check	Result
Core allocation	The current F2R + ADAA governed strategic allocation policy was confirmed. Exact provider composition remains private.
Historical Core	The displayed history applies the current Core policy consistently throughout; earlier allocation policies are not spliced into the series.
Rebalance timing	The monthly decision and rebalance dates matched the timing rules described in Section 5.
Transaction costs	Drift-based turnover and the 10 bp-per-side cost assumption matched the operating calculation.
Performance measurement	The daily return series, 252-trading-day annualization, and drawdown calculations reconciled.
Data coverage at release	The common daily history ran from 1 May 2017 through 31 August 2026, covering 2,347 observations.
Current Mark	The Current Mark remained a monitoring view and did not alter the Official Decision or completed history.
Display consistency	The Core, Forward Preview, Forward Shadow, and Dynamic FX labels matched the documented role of each view.

What changes over time

The validation table is dated to this release. Later monthly data extend the historical record; a new document version is needed only when the methodology or portfolio rules change materially.

RELATED METHODS

14. Related Quantitative Methods

The SlackQuant Quantitative Methods library provides background on the techniques referenced in this document. The entries below are the methods most directly related to PDS.

ID	Method	Relevance to PDS
QM007	Portfolio Backtesting and Rebalancing	Explains monthly rebalancing, weight drift, and next-period return timing.
QM008	Drawdown, Underwater Duration, and Recovery	Covers drawdown depth, duration, recovery, and running peaks.
QM009	Turnover, Transaction Costs, and Net Performance	Explains turnover, transaction costs, and net portfolio returns.
QM011	Strategy Similarity and Decision Diversification	Provides the basis for comparing strategy decisions rather than return correlation alone.
QM013	Benchmark-Relative Portfolio Evaluation	Covers active return, tracking error, information ratio, relative wealth, and relative drawdown.
QM014	Information Timing, Data Vintages, and Real-Time Macro Data	Explains release timing, data vintages, revisions, and the information available at each decision date.

Methods library: research.slackquant.com/methods/

How these references are used

The Methods library provides general background. This document explains how those ideas are applied in PDS, especially where timing, transaction costs, or portfolio status affect interpretation.

LIMITATIONS

15. Limitations and Revision Policy

Limitations

- Historical results and observations from live operation do not guarantee future returns.
- The current governed strategic allocation between F2R and ADAA is the present PDS policy; its exact provider composition is private and it is not presented as universally optimal or permanent.
- F2R and ADAA can still share market exposures even though their decision processes differ. Their combination should therefore be evaluated from the observed portfolio results rather than assumed to be diversified by design.
- Transaction costs are modeled assumptions. Actual trading costs can differ.
- The historical Dynamic FX comparison is not evidence of realized forward/NDF hedging performance.
- Forward Preview and Forward Shadow are monitoring tools and do not change the Official portfolio unless they are separately adopted through the regular decision process.
- This document describes the PDS portfolio framework. Broker fills, custody records, account holdings, and trade confirmations are outside its scope.

When a new version is needed

Change	Update needed
Presentation-only edit	Normally no new version unless the meaning changes.
Routine data update or new completed month	No new version; the historical record is simply extended.
Core strategy membership or target allocation	New document version.
Rebalance timing, turnover, or transaction-cost treatment	New document version.
Definition or role of Official Decision, Forward Preview, or Forward Shadow	New document version.
Public disclosure policy	New document version.
FX overlay adopted as an implementable portfolio component	New document version.

Versioning rule

A new version is issued when portfolio construction, timing, cost treatment, decision status, or disclosure policy changes materially. Routine monthly data updates do not require a new version.

Appendix A. Decision and Publication Status

View	Current?	Can change the Official target?	Performance?	Published?
Official Decision	Yes	Yes - through the regular monthly decision process	Realized after the scheduled rebalance	Current target: no; delayed history: yes
Current Mark	Yes - incomplete period	No	Provisional current-period view	No
Forward Preview	Yes - incomplete period	No	No Forward Preview performance	No
Forward Shadow	Yes - monitoring	No	Separate comparison performance may be tracked	No
Public Historical Release	Dated	No	Completed historical performance	Yes
Historical Dynamic FX comparison	Dated	No	Historical comparison only	Yes, when released

Reading the table

- Only the Official Decision can change the portfolio target. Forward Preview, Forward Shadow, and Current Mark are monitoring views; public releases are delayed historical records.

Appendix B. PDS Resources

Resource	Purpose	Location
PDS System page	Overview of the system, current Core policy, and related public resources.	research.slackquant.com/systems/pds/
PDS Dashboard	Delayed historical performance and portfolio information for completed periods.	research.slackquant.com/systems/pds/dashboard/
PDS System Documentation	Detailed explanation of portfolio construction, timing, monitoring, and disclosure.	Linked from the PDS System page
Quantitative Methods	Background on the quantitative methods referenced in this document.	research.slackquant.com/methods/
Research	Separate empirical papers and research records related to the strategies and methods.	research.slackquant.com/research/

Start with the System page for an overview. The Dashboard provides delayed historical results; this document and the Methods library provide the supporting detail.

Appendix C. PDS v1.1 Summary

Item	PDS v1.1
System	Portfolio Decision System (PDS)
System type	Portfolio Decision & Operating System
Current strategies	F2R and ADAA
Core allocation	Governed strategic allocation integrating F2R + ADAA; exact provider composition private.
Rebalance frequency	Monthly
Between rebalances	Provider and asset weights drift with market returns; there is no daily reset to strategic provider targets.
Rebalance timing	First U.S. trading-day close available to both strategies after the signal month ends.
Turnover measure	One-way turnover is half the sum of the absolute changes between pre-rebalance and target weights.
Transaction-cost assumption	10 basis points per side.
Performance calculation	One continuous daily return series; monthly and annual results are derived from it.
Forward Preview	Monitoring only; not executed and no Forward Preview performance is reported.
Forward Shadow	Separate comparison to the Active Core; not part of the Official portfolio unless adopted.
Public disclosure	Delayed historical information; the current private portfolio state is not disclosed.
Dynamic FX comparison	Delayed historical comparison only; not the current FX position.
Implementable FX overlay	Not part of the current PDS portfolio. An implementable version would require tradable forward/NDF data, roll rules, hedge turnover, and transaction costs.

About this version

This version describes the PDS framework in effect on 11 September 2026. Earlier allocation policies and development versions remain part of the project history, but they are not used to describe the current system.

SlackQuant Research

System documentation only. Not investment advice.